Date	16 July 2026
Project Name	Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction
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Maximum Marks	3 Marks

DESIGN THINKING: EMPATHY MAPPING

To ensure that the **Smart Lender AI** platform addresses real-world frustrations rather than just solving a technical classification problem, a Design Thinking approach was utilized. This Empathy Map focuses on our primary user persona: **The Loan Applicant**, a modern individual seeking quick, transparent, and hassle-free credit options. By analyzing what the applicant says, thinks, does, and feels, we can directly map system features—such as instant predictive pipelines and real-time explanation metrics—to actual user anxieties and desires.

SAYS (VERBALIZATIONS)

- "Why does a loan decision have to take weeks?"
- "What exact documents do I need to prove my eligibility?"
- "Can someone explain why my application was rejected?"

THINKS (INTERNAL BELIEFS)

- "Is my poor credit history going to instantly disqualify me?"
- "I hope they don't lose my paper application in a stack somewhere."
- "Is this system evaluating me fairly or using biased data?"

DOES (BEHAVIORS)

- Manually gathers bank statements and tax forms.
- Constantly refreshes bank portals or calls agents for status updates.
- Uses online calculators to guess their approval odds before applying.

FEELS (EMOTIONAL STATE)

- **Anxious** about credit inquiries hurting their score.
- **Overwhelmed** by complex financial jargon.
- **Frustrated** by a lack of communication from traditional banks.

Pains vs. Gains Summary

Understanding these friction points directly influenced the machine learning model's priority features:

The Pain Points

Traditional manual underwriting creates an agonizing "black box" where applicants have no visibility into how heavily criteria like credit history vs. applicant income affect their chances. This breeds confusion, systemic mistrust, and emotional exhaustion.

The Gain Realizations

Smart Lender AI bridges this gap. By knowing that the underlying system weights **Credit History** at 46% and **Co-applicant Income** at 15%, credit officers can confidently explain system decisions, turning an intimidating financial hurdle into a transparent, user-friendly experience.